# Everything but the Software:

## Feature-Driven Decay in an Agentic Coding Loop. An Exploratory Case Study and an Experiment Protocol

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Draft of 1 November 2026

**Abstract**

Each feature changes the starting point for the next. In agentic software development, that starting point includes code, tests and the instructions guiding later work. This paper examines how faults and corrective rules accumulate in that inherited state. Maintenance literature and agent benchmarks distinguish conversational errors from repository-carried failures. A practitioner-led demonstrator supplies a motivating case: agent-assisted research, business and product requirements, architecture decisions, Design Thinking critique and disposable prototypes informed epics and user stories. Despite preparation intended to improve delivery speed and quality, extended autonomous work left the replacement unusable for the participant and prompted renewed feedback and refactoring. Five candidate mechanisms organise this transition: specification inflation, proxy capture, learning evaporation, process-gate inflation and feedback starvation. Migration and deployment dependencies remain central to the observed failure. A prospective protocol specifies 495 trajectories across three selected models and five arms: feature-only delivery, scheduled consolidation, read-only review, matched-resource delivery and visible regression repair. The outcomes are final behavioural coverage, retained-contract loss, an unforeseen change request and resource cost. The design targets a ten-percentage-point coverage gain under an explicit variance assumption; consolidation combines structural work, additional tests and a revised handover. Source freeze, prospective registration, affordability and collection authorization remain gates. The study asks whether consolidation improves the system inherited by later feature work.

**Keywords:** agentic software engineering; coding agents; spec-driven development; software evolution; technical debt; refactoring; feedback loops; human–AI interaction; exploratory case study; experiment protocol.

## 1 Introduction

On 1 August 2012, Knight Capital's automated router sent more than four million orders while attempting to fill 212 customer orders. The firm lost more than USD 460 million. The SEC traced the incident to a defective function left in the router after a 2005 change, an incorrect deployment for a new trading programme, and inadequate controls. The new deployment activated old code that was no longer meant to run [87]. The failure crossed a feature boundary and a seven-year interval: the relevant starting point was the accumulated system, not just the latest patch.

Less spectacular versions of the same maintenance problem are familiar. Cunningham's 1992 account of WyCash described a financial product continually adapted to market demands. Its mature code was repeatedly revised and rewritten; the technical-debt metaphor explained why

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shipping a first-time implementation required later rewriting as understanding improved [26]. Some debt is visible as duplication or awkward dependencies. Some remains hidden in assumptions about inputs, deployment and earlier behaviour until another change depends on it. Counting completed changes reveals neither.

Agentic development makes this distinction newly pressing. PhotoCraft, an early-alpha reimplementation of Photoshop, reports 631 of 631 menu items live, but explicitly defines that count as dispatch rather than behaviour. Its own dated assessment estimates about 76% feature breadth and 45% readiness for real work [75]. A command can be reachable without the surrounding workflow being dependable. At the repository level, SWE-Maintain finds that agents solve fewer later tasks when starting from their own earlier changes than from the developers' changes, even though each task starts in a fresh session [69]. The next session can forget the conversation and still inherit its mistakes.

The engineering response has a long history: isolation and explicit contracts limit dependencies; tests and continuous integration expose regressions; small releases shorten feedback; refactoring preserves room for later change [7, 8, 31, 71]. Recent discussions by Beck, Martin and Fowler revisit these concerns for agent-assisted work [15, 32, 58]. Beck's distinction between producing code and learning through working software sharpens the research question [13, 15]: what does a feature leave behind for the next one?

The question arose in an architectural demonstrator developed by one practitioner with multiple agents. Research documents, business and product requirements and architecture decisions were built through sustained conversation, challenge and owner decisions. Design Thinking methods, including what-if scenarios, persona analysis and journey mapping, helped question the proposed solution; disposable pilots and prototypes exposed workflow problems before a usable MVP emerged. A backlog agent then converted that accumulated context into epics and user stories for a delivery coordinator. The intended advantage was faster implementation of better-understood features, supported by explicit acceptance criteria and review.

Extended autonomous sessions nevertheless left the replacement unusable for the participant. Renewed product-use feedback redirected implementation and refactoring. This sequence motivates investigating how preparation and local completion checks translate into the behaviour inherited by later work. The repository, pull requests, append-only log and versioned coordinator instructions retain the evolving delivery record [80]; the source package pins the evidence used here. The repository may be temporarily unavailable during the experiment embargo, after which the approved history and context are intended for release. The author's earlier experience with Aithena, a book-search project with PDF ingestion and a browser interface discussed separately in the supplement, and the PhotoCraft inspection selected after the motivating-case analysis have separate provenance in the ancillary anc/supplementary-context.md.

The working hypothesis is that feature-driven delivery can carry unresolved errors and avoidable change burden forward unless it includes cumulative verification and consolidation. Two kinds of state matter: the software being changed, and the instructions used to change it. Adding a corrective rule after every failure can leave prompts, skills and handover notes with their own maintenance burden. A fresh agent inherits both kinds of state. Section 2 distinguishes these pathways; the case then examines how they interact with completion proxies, review capacity and product-use feedback.

Here *feature-driven decay* describes loss of retained behaviour or increasing change burden across feature steps. Whether feature-driven decay occurs remains a question for the prospective study.

Three questions guide the investigation:

- **RQ1.** Through which retained state can errors and change burden carry from one feature to the next, and what does existing evidence show about those pathways?
- **RQ2.** How does an exploratory delivery record illuminate that problem, and which candidate mechanisms and rival explanations does the gap between completion signals and usable

replacement behaviour suggest?

- **RQ3.** Does scheduled agent-led consolidation after each feature improve retained behaviour and future changeability, and at what cost?

The report makes three linked contributions:

1. A focused synthesis distinguishing conversational self-conditioning, repository-carried faults and instruction maintenance, with a replot of published paired-history data (Section 2).
2. An exploratory account of the transition from agent-assisted discovery and challenged requirements to coordinated feature delivery and corrective product feedback. Five candidate mechanisms and within-case contrasts connect the interruption of usability to competing explanations (Sections 3–5).
3. A consolidation harness and a prospectively sized five-arm protocol to estimate retained behaviour, future changeability and resource cost. It follows the registered-report example of Lulla et al. [55] and HVE Core's Minimum Viable Experiment method (Sections 6 and 7).

The case concerns a migration and a configuration designed by the author, including delegated delivery and merge decisions. The case suggests hypotheses whose causal relationships remain to be tested. The planned pilot estimates a multi-component, extra-resource package comparison; it does not isolate refactoring or validate the five case mechanisms individually. Registration and launch checks precede collection. The case's validity and the pilot's identification limits are consolidated in Sections 5.2 and 7.3.

# 2 Background

## 2.1 Accumulation and inherited state

Lehman's laws and Parnas's software-aging account motivate maintenance alongside growth [51, 52, 71]. The 1997 formulation states the declining-quality law; it supplies no data for or against it. Eick et al. report mixed evidence from fifteen-plus years of change history [29], and a systematic review finds uneven support across systems [40]. Feature count and churn describe activity. A regression loses still-required, previously working behaviour; an unimplemented capability is an acquisition failure. A failure that continues across changes is persistence. Establishing that it caused a distinct later defect requires a trace of that dependency.

Cunningham describes shipping first-time code and rewriting it as understanding improves [26]. Fowler's design stamina is explicitly a hypothesis [31]. Reported debt costs concern different settings: a vendor-associated 39-codebase study reports 124% longer issue resolution in low-quality files [86]; 43 developers self-report about 23% of working time lost to debt [16]. These are not universal productivity coefficients.

**What survives the next session?** Code, tests and deployment assumptions survive through the repository; instructions and handovers survive through procedural context. Either channel can preserve a sound decision or a mistake. Here "software entropy" provides an engineering analogy for the maintenance problem. The proposed measurements concern retained behaviour, adaptation and resource use.

## 2.2 Evidence within and between features

Conversational benchmarks identify dependence on erroneous histories, not merely context length. Zhang et al. study false justifications after wrong answers [93]; Sinha et al. vary erroneous prior state updates in a controlled execution task [83, Secs. 2.2–3, Figs. 5–6]. Laban

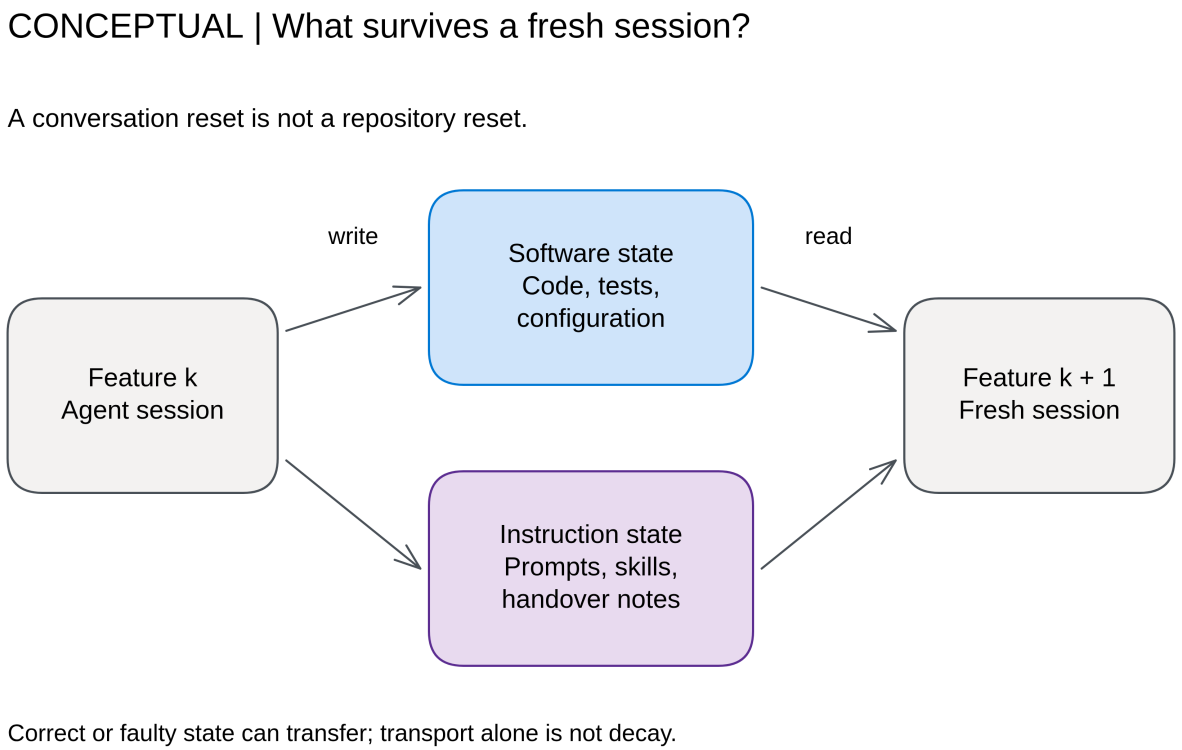



**Figure 1:** Conceptual read/write paths between sessions. A fresh conversation inherits software and instructions. Arrows identify state-carrying paths whose effects require measurement.

et al.'s 39% relative multi-turn performance drop concerns simulated underspecified conversations, not repository defects; locators refer to their arXiv v1 [49, Secs. 3.3, 6.1, App. F].

Pan et al.'s SWE-Maintain more directly tests repository inheritance: 100 three-event episodes from 50 Python repositories, with fresh sessions and matched later tasks under developer and agent histories. Across 1,996 paired later events, agent history reduces current-task resolution by 5.6 points; requiring earlier tests also to pass gives a 14.0-point gap. Unweighted mean current-task gaps grow from 4.0 to 7.2 points between the second and third events. In the selected 619-event clean-history subset, the current-task difference is $-1.1$ points, interval $[-3.8, +1.6]$, and cumulative event resolution differs by $+2.7$, interval $[-0.5, +6.1]$ [69, Secs. 4.1–4.2, Tables 2–3]. This favours inherited behavioural failures over complexity alone as the main observed channel, without establishing a general decay law or a randomised repair effect.

Their `python_unidiff` example reuses line-splitting logic that loses terminators in a later constructor; Figure 1's caption introduces it and Section 5.1/Figure 5 supplies the detail [69]. SlopCodeBench and SWE-CI extend iterative evaluation [23, 68]. Borg et al.'s maintainability study finds no significant downstream difference for late-2024 assistants; only phase two was randomised [19]. Its findings provide a useful contrast across tool generations and collaboration settings.

## 2.3 Instruction maintenance and evaluation

Instruction files themselves require maintenance [67, 76]. Mohsenimofidi et al.'s 169 commits have 242 overlapping assignments across nine categories; 78 instruction additions, 59 modifications and 23 removals are three categories, not a partition [63, Sec. 4.3, Table 2]. Chatlatanagulchai et al.'s sampled maintenance labels likewise include addition and pruning [21]. Instruction debt here is the author's term for obsolete, duplicated, conflicting or poorly routed directives.

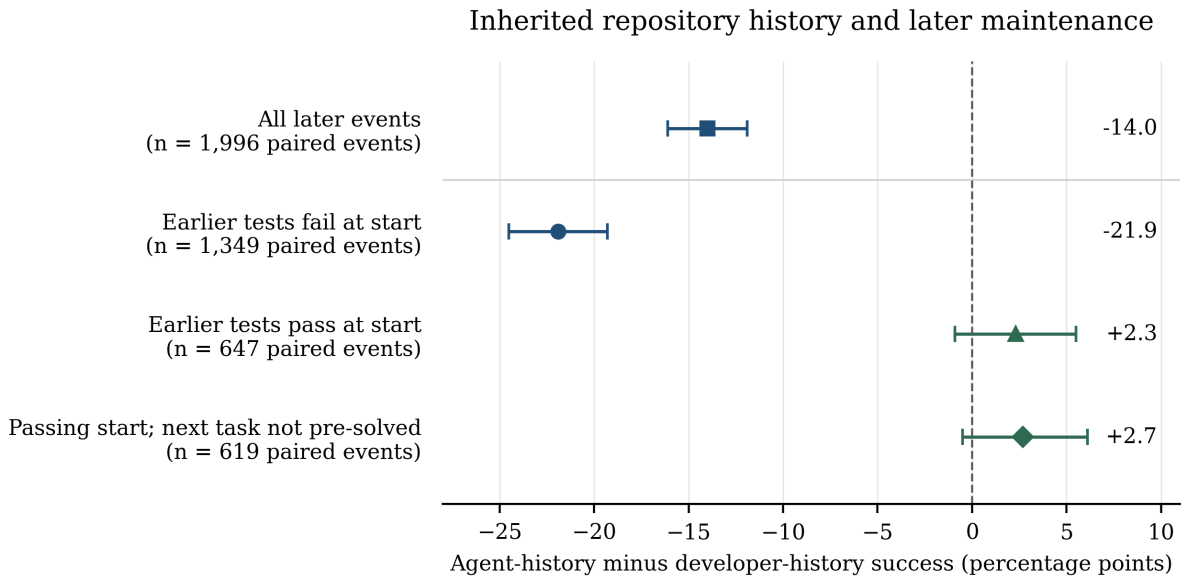



**Figure 2:** Published event-resolution differences and repository-clustered 95% intervals from Pan et al., Table 3. Current and prior non-superseded tests must pass. The overlapping selected cohorts apply different restrictions to the same benchmark. Points and whiskers reproduce the published estimates and intervals.

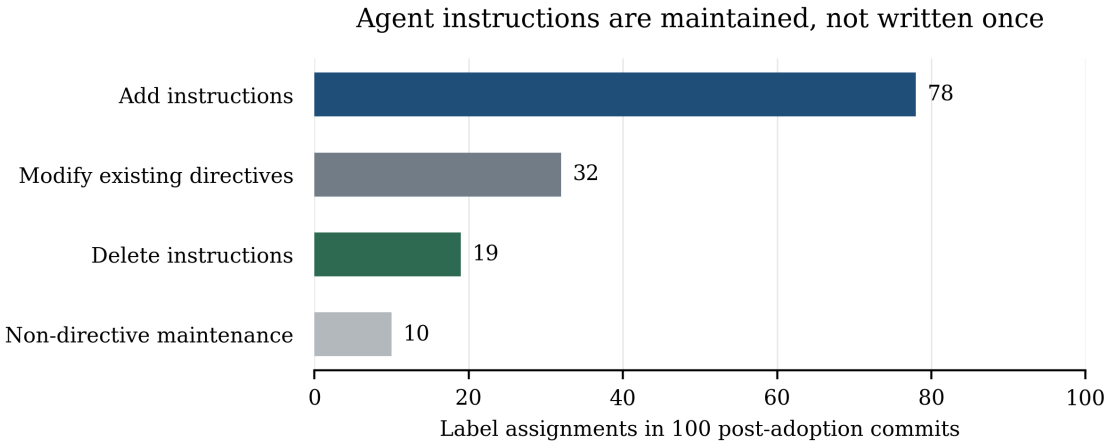



**Figure 3:** Labels in 100 sampled post-adoption commits, from Chatlatanagulchai et al., Sections 4.2.2–4.2.3. Labels overlap and count edit types, not net token growth or effects on agent success.

Gloaguen et al. report 20%/23% higher inference cost with LLM-written context and slightly lower success, by 0.5/2 points, without significant differences from no-file baselines. Developer-written files outperform LLM-written ones on CTXbench ($p = 0.038$) [36, Sec. 4.2, Tables 2–3]. SkillOpt and Jiang et al. provide counterevidence to treating retained instructions as inherently harmful: evaluated procedural guidance can help, while retrieval and downstream success remain distinct [44, 62].

**Tests and review.** RefactorBench studies refactoring, not between-feature consolidation [35]; developer-agent collaboration studies are observational [48]. SWE-Sharp-Bench supplies reproducible repository evaluation [59], while CRBugBench exposes review limits [90]. Building to the Test finds a dead or absent required library in seven of twelve oracle-in-the-loop runs [56]. ImpossibleBench tests exploitation more broadly [94]. Test integrity and adequate outcome

coverage are different requirements. AgentRx motivates traceable failure diagnosis [60].

**Bounded correction.** Pavlenko et al.'s factory optimises human-architected systems against measurable objectives; productivity estimates cover application work, excluding infrastructure and onboarding [72]. AutoSaddler treats harness updates as offline learning [70]. Neither isolates the present intervention and later change probe.

**Reflection and self-improvement.** Ng's reflection design-pattern letter explicitly recommends Self-Refine, Reflexion and CRITIC [66]; he is not a coauthor of those papers. Self-Refine iterates feedback and revision with the same model without training [57, Sec. 2]. Reflexion retains linguistic feedback in episodic memory for subsequent trials rather than updating model weights [81, Sec. 3]. CRITIC uses external tools to check and revise an output [38, Sec. 3]. These are precedents for reflective correction, not evidence that every retained lesson or revision improves later feature work. Their feedback and repeated-trial budgets differ from the present protocol, whose hidden evaluator outputs never reach agents.

Ng coauthored MedAgentBench, which evaluates task completion in an interactive medical-record environment rather than chatbot answers alone [43, arXiv v2, Secs. 2.3–2.4]. Its evaluation of observable action outcomes informs the measurement focus of this protocol.

The two owner-selected videos illustrate these boundaries rather than provide experimental findings. In the Ng/Moroney talk, Ng describes human-feedback iteration; Moroney later warns about code structure and describes task-level reflection [74]. In the separate memory presentation, a speaker identifying herself as Anthropic staff describes persistent memory and asynchronous transcript-driven curation, with human review of proposed updates [5]. Both sources are third-party uploads checked through automatic captions, not audio-verified research reports. Neither supplies a controlled cross-feature consolidation result.

**Engineering rationale.** Beck's alternation of features and tidying aims to restore future options [10–12, 15]. Extending that metaphor to acceptance criteria is this author's interpretation. Understanding/abstraction and learning through use have separate sources [14, 15]. Tests, short releases, explicit design responsibilities and usable increments motivate the checklist [6–8, 20, 24, 33, 34, 42, 58, 65, 78]. Monitoring burden and completion proxies qualify the rationale [2, 25, 37, 50, 82, 84, 85]. Spec-driven tools structure handoffs; the intended outcomes still need execution checks [1, 18, 27, 88].

**HVE Core.** Microsoft's Hypervelocity Engineering Core supplies file-based Research–Plan–Implement–Review handovers; the author's coordinator added autonomous dispatch. The reported case snapshot is `plugins-v3.3.106`, not proof of its installation date [61]. Additional source qualifications and reproducibility detail are in the supplement.

# 3 Case and method

## 3.1 Setting

The motivating project was an architectural training exercise developed by one practitioner acting as architect, developer and product owner. Its synthetic, multi-role service workflow combined documentary evidence, approvals, identity, data and external-service dependencies. The exercise required research, requirements, architecture, a delivery plan and a working demonstrator. These interdependencies made preservation of a complete user journey consequential even in a non-production system. No operational service or real resident data is evaluated here.

Agents performed most research synthesis, drafting and coding, with the practitioner steering scope and decisions. Several weeks of discovery and requirements work preceded roughly a

week of interactive MVP development and a ten-day autonomous delivery window. The project continued afterward; its current state and detailed chronology belong to the versioned repository record [80]. An automated repository scorer supplied an additional completion signal. Internal assessment wording is not reproduced.

### 3.2 Workflow and tools

GitHub Copilot and HVE Core supported a sequence of connected activities:

- Research agents assembled a knowledge base of source snapshots, maintained syntheses, an index, owner-curated memory and an append-only log.
- The Design Thinking coach and stakeholder-role agents challenged assumptions through what-if scenarios, persona analysis and journey mapping. Their critiques separated bounded AI tasks from deterministic controls and accountable human decisions. Role-agent answers were hypotheses for the exercise; direct stakeholder research remained a separate requirement.
- BRD and PRD builders, quality review and the ADR assistant turned successive discussions and owner decisions into revised requirements and architecture records. These documents changed as understanding developed.
- Disposable pilots and prototypes tested candidate workflows. Their critiques and participant feedback informed a deployed MVP that the author used intermittently and refined through short feedback loops.
- A backlog agent converted the accumulated context into epics, features and user stories, with acceptance criteria, dependencies and indications of existing MVP coverage. Some stories also exposed scope needing subsequent requirements reconciliation.
- A custom coordinator then dispatched implementation and review work against that backlog (Section 3.3).

The intended benefit was accelerated planning and better-grounded feature delivery. Neither a human-only comparator nor a measured speed or quality gain was available. The retained documents and conversations matter because they shaped what workers were asked to build and reviewers were asked to accept.

### 3.3 The coordinator

The coordinator ran through Copilot CLI on a VM. It selected dependency-ready issues and launched parallel worker sessions in isolated git worktrees. Each worker received an issue and HVE Core's research, plan, implement and review (RPI) workflow. The coordinator checked the resulting pull request, updated the project board and, under owner delegation, merged changes. Interactive and CLI model configurations changed during the project; recorded selectors and recalled interface labels are distinguished in the source package. The delivery record is consequently an evolving workflow.

Early worker sessions implemented directly, prompting the author to make the full RPI sequence and committed tracking artefacts mandatory. Further instructions required evidence-backed acceptance ticks, scheduled monitoring and increasingly comprehensive local gates. Human decisions, approvals, credentials and unresolved review findings could still block work. At the same transition, an accepted architecture decision prioritised migration to a new source layout and froze feature work in the MVP. Coordination, migration and procedural requirements therefore changed together.

### 3.4 Data and analysis

This exploratory single-case study follows the transition from interactive discovery and MVP use to backlog-mediated autonomous delivery. The author selected the case after encountering the unusable replacement and acts as participant-observer. Case-study guidance and the ACM SIGSOFT Empirical Standards inform the account [77, 79].

The original analysis pins repository commit `29c6b51`; a separately pinned follow-up uses `9b74ac8d`. Sources include the knowledge-base log and index, requirements, architecture decisions, coordinator definition, issues, pull requests and retained RPI records. The author's review adds process clarification and an export of project issue states, retained as anc/project-insight s-2026-11-01.csv. Source dates, historical counts and extraction rules remain in the ancillary records. The main account uses them to reconstruct the sequence and handoffs.

Analysis combined descriptive measures, reference tracing of recorded lessons and AI-assisted interpretation of selected episodes. It did not include systematic entry-by-entry coding. Early log entries were reconstructed from history; later entries are contemporaneous project accounts. Specification review was selective. Participant recollection, committed source and execution evidence remain distinct, including in the author's fuller account of renewed feedback. The original source package retains missing score-to-release and deployed-journey links.

External repositories and literature were selected for relevance to inherited faults, instruction maintenance and outcome verification. They supply contextual comparisons and a focused narrative synthesis. The published-evidence figures reproduce reported values with source locators; the external applications and original studies were not rerun. The ancillary observation record preserves its purposive issue sample, exclusions and retrieval boundaries. These sources are analysed separately from the motivating project.

### 3.5 Constructs and evidential boundaries

Here an *agent* is a configured model–tool–instruction loop that selects and executes actions, not the model alone or an independently accountable person. A *skill* is a reusable procedural resource describing when and how to act and verify the outcome, distinct from an execution history or a model-weight update. This procedural distinction follows Jiang et al. [44, Sec. 1]; the pilot's handover note is retained context, not a separately manipulated skill treatment.

*Working behaviour* means completing an intended task through the running system, including the identity, data and service dependencies needed for that task. A healthy endpoint or a passing repository test alone does not establish it. *Decay* denotes loss of previously working behaviour or increasing difficulty of understanding and changing a system. The case records lost usability during replacement; any decline in intrinsic code quality remains unmeasured. Deployment drift could produce the same observation. *Adaptability* is operationalised prospectively as success and cost on the fixed-budget, unforeseen change request, not as a universal measure of maintainability. *Consolidation* means scheduled structural work intended to preserve observable behaviour, with tests checking that intention; passing finite tests cannot prove semantic equivalence.

Five measurement categories remain separate. PRs, lines and words describe *production volume*; checks, reviews and receipts describe *validation activity*; observed user tasks describe *system behaviour*; code metrics and the change request probe *internal quality and changeability*; and corrective help requested or supplied describes *intervention*. The original observations cover the first three. A later post-hoc comparison adds structural proxies, not a controlled changeability task or systematically coded intervention dataset. Neither volume nor validation activity is used as a direct estimate of decay.

### 3.6 Chain of evidence and reflexivity

Table 1 separates observations, interpretations and proposed checks. The relevant evidence concerns how discovery was carried into implementation, how completion was judged and how

product feedback returned. Repository sources refer to the original case pin unless a later record or participant account is identified. Detailed locators and unresolved execution links remain in anc/case-evidence-matrix.json.

**Table 1:** Evidence relevant to the delivery transition and checks that would distinguish the candidate explanations.

| ID | Observation and provenance | Interpretation | Discriminating evidence |
|---|---|---|---|
| O1 | The log and participant account describe a used MVP followed by a replacement lacking identity, staff and model prerequisites. | Delivery failed to preserve a complete working journey. | Execute the same journey against versioned deployments and configurations. |
| O2 | Pull requests and automated assessments recorded completion while participant sign-in remained blocked. | The completion signals covered a different or insufficiently linked outcome. | Link the assessed capability and release to the failed deployed task. |
| O3 | Requirements and process records expanded through discovery and delivery; even a small source correction carried a substantial evidence record. | Documented context may outgrow what a worker or reviewer can use effectively. | Compare actual context access and review effort with validated behaviour. |
| O4 | Simplicity and deployment lessons were recorded, but later replacement work omitted previously working prerequisites. | Lesson availability and application may diverge. | Trace the relevant session inputs and whether the lesson was actionable or deliberately deferred. |
| O5 | Coordinator rules formalised gates and delegated merging; the author continued resolving blockers and later requested stabilization. | Approval activity and product-use feedback served different functions. | Compare outcome checks, steering decisions and subsequent implementation changes. |

The author selected the episodes, designed the coordinator and had an interest in successful delivery. Those roles favour a coherent retrospective explanation. Experience-based beliefs can differ from project evidence [28]; the same caution applies to AI-written records and AI-assisted synthesis. Independent checking of O1–O5 against retained source and execution records remains outstanding. Jiang et al. make mechanism analysis inspectable through matched trajectories, evidence-linked labels and a separate human check of taxonomy construction, while distinguishing that check from exhaustive human coding [44, Sec. 4, App. A.2]. Here the five mechanisms organise selected episodes. Their prevalence and causal effects require a different design. The evidence index retains null release or execution identifiers where a link still needs checking.

## 4 Exploratory testbed: observations

The relevant sequence is the movement from exploratory dialogue to a formal backlog, autonomous issue delivery and renewed product feedback. Table 2 summarises those phases. The detailed project chronology and changing totals remain in the repository and source package.

**Table 2:** Phases of the motivating case and the state carried forward.

| Phase | Activity | State inherited by later work |
|---|---|---|
| Discovery, several weeks | Research, role-agent and Design Thinking challenge, what-if scenarios, persona and journey analysis; iterative BRD, PRD and ADR work. | A rich but evolving requirements and decision context. |
| Interactive probes, about a week | Disposable prototypes, participant critique and short implementation/feedback cycles produce a used MVP. | Working journeys, simplicity lessons and deployment prerequisites. |
| Backlog handoff | A backlog agent derives epics and user stories, criteria, dependencies and existing-coverage labels. | A formal work hierarchy that selects implementation and review tasks. |
| Autonomous delivery, about ten days | Parallel RPI workers deliver issues under a coordinator while migration and gate requirements evolve. | Merged changes and validation records; the replacement lacks a complete usable journey. |
| Renewed steering and follow-up | The participant prioritises usable increments, functional checks, refinement and refactoring; stabilization is formalised. | Product-use feedback again shapes what subsequent work must establish. |

### 4.1 From discovery to epics and user stories

Preparation involved sustained steering rather than a single specification prompt. Role-agent challenges questioned the requested end-to-end agents; what-if scenarios tested alternative inputs and responsibilities. Persona and journey work exposed handoffs, missing information and approval boundaries. The resulting BRDs, PRDs and ADRs were critiqued and revised through further conversation. These activities changed the proposed solution before and during implementation.

Disposable prototypes then made some assumptions tangible. Participant critique and an agent-assisted simplicity review produced requirements for clear ownership, a visible next action, understandable blockers and manual paths that remained usable without AI. Short feedback cycles reshaped an MVP around those lessons. The author reports using it intermittently, including sign-in, synthetic data and model-backed assistant paths.

The backlog manager drew on this accumulated context to generate the work hierarchy, acceptance criteria and dependencies. Existing-coverage labels distinguished proposed work from functionality already present in the MVP. The supplied project export records a batch expansion around this handoff, followed by movement into completed states. It provides a trace of work organisation; accepted user journeys require their own evidence. Requirements reconciliation continued afterward, including for scope first surfaced in backlog stories. Thus preparation and delivery overlapped, and the issue hierarchy remained an evolving interpretation of the design.

### 4.2 Local completion and the usable workflow

Autonomous delivery changed the feedback path. The coordinator selected issues and workers satisfied local implementation and review obligations. At the same time, a source-layout migration replaced the used MVP. The replacement passed bounded engineering checks, including health, restart and unauthenticated-request rejection, but lacked identity, staff and model prerequisites needed for the participant's complete workflow. Restoring a public entry point left sign-in dependencies unresolved.

The author continued supplying approvals and resolving technical blockers. What diminished

was the ability to exercise realistic work in the replacement and use that experience to steer the next change. Missing prerequisites became further issues, each with research, planning, review and correction work. Scope additions and validation costs also changed delivery conditions. These observations locate a concrete interruption of usable behaviour within a broader process of accumulating obligations. The automated score and failed journey still need release-and-scope linkage before their disagreement can be assessed as a matched comparison.

### 4.3 Bounded within-case contrasts

Table 3 compares episodes and negative cases already in the record. Migration, models, workload and gate rules changed alongside coordination, so these contrasts motivate explanations for later testing.

**Table 3:** Contrasts relevant to context, completion and feedback.

| Contrast | Recorded difference | Analytical consequence |
|---|---|---|
| Interactive MVP versus autonomous replacement | Product use supported short feedback cycles; the replacement could not support the same task. | Feedback is a candidate explanation alongside the changed migration and environment. |
| Requirements before versus during autonomy | Expansion and review obligations already existed during discovery. | The burden has a longer history than the coordinator, and some detail preserves important controls. |
| Engineering checks versus participant task | Bounded checks passed while real-user sign-in remained blocked. | Acceptance needs coverage of the whole journey and its deployment dependencies. |
| Exploratory challenge versus backlog execution | Role agents questioned the initial proposal; delivery workers implemented specified stories. | Agent behaviour depended on task framing, context and opportunities for challenge. |

### 4.4 Renewed feedback and follow-up

The participant requested a usable increment, prioritised feedback-first work and asked the coordinator to pause, exercise the accumulated functionality and refactor. The participant's account places this renewed feedback at the turning point in making the replacement useful. Subsequent source history records implementation refinement and recurring stabilization checkpoints. The retained checkpoint review qualifies its local engineering evidence and leaves live acceptance separate.

This follow-up matters because the project was steered through interaction: the backlog and the worker instructions did not settle the product outcome once and for all. A dated, release-linked journey is still needed to measure time to restored usability. The original observation window and the later repository snapshots remain distinct. The repository supplies the evolving history; the ancillary records preserve the exact observations used here, including changing issue totals, document volume, lexical counts and static source metrics. Those records support reconstruction while the prospective experiment supplies a separate test of retained behaviour and changeability.

## 5 Five candidate mechanisms

Five candidate mechanisms organise the gap between extensive preparation, local completion signals and an interrupted usable workflow. Table 4 summarises the indicators; the supplementary operational audits set out definitions, counter-evidence, rivals and discriminating checks. Together they connect the inherited-state problem to the way this delivery loop selected work, carried lessons and checked outcomes.

**Table 4:** Five candidate mechanisms, case indicators, relevant prior work and proposed early-warning signals. Indicators are not causal tests.

| Candidate | Case indicators | Related prior work | Early-warning signal |
|---|---|---|---|
| Specification inflation | Expanding requirements and acceptance obligations during both discovery and implementation | Royce; Gall; Brooks; Böckeler; Beck | Specification growth versus human-validated behaviour |
| Proxy capture | Artefacts became a merge condition; completion assessments coexisted with blocked sign-in, with release/scope linkage still pending | Goodhart; Strathern; Beck's genie; ImpossibleBench; METR | Artefacts per behaviour change; scores that disagree with a live demo |
| Learning evaporation | Recorded simplicity and deployment lessons were followed by missing prerequisites; actual session access remains unknown | Naur; context studies; multi-agent failure taxonomy | Lessons that exist only as prose; problems solved twice |
| Process-gate inflation | Broad per-push gates, constrained validation resources and an exception for a small correction | Bainbridge; ironies of generative AI; Murphy; Goldratt | Gate time per change; review minutes per merge |
| Feedback starvation | A used MVP was frozen; the participant could not exercise the replacement and requested usable increments | XP; incremental delivery; tracer bullets; Agile Manifesto; DORA | Time to first human use; age of last usable increment |

They occupy different points in the delivery process (Table 5). Specification growth increases the obligations and interactions that later work must preserve. Learning evaporation concerns the path by which a relevant lesson reaches the next session. Proxy capture concerns what selects and approves work; process-gate inflation and feedback starvation concern the capacity to detect and correct cumulative problems. None needs to be called error propagation to explain how an inherited fault can survive.

Before the participant's checkpoint request, the loop had no scheduled cumulative consolidation step. Its large restructuring, the source-layout migration, was planned as backlog items; local refactoring could still occur within individual issues. The experiment separates retained-contract loss, structural fidelity diagnostics and later adaptation. Formal fault-lineage annotation would address the stronger question of whether an earlier fault generates distinct later failures; it is deferred rather than added to the pilot's workload.

**Table 5:** Accumulation, transmission and correction in the working model.

| Process | Question | In this case |
|---|---|---|
| Accumulation | What obligations and change burden does each feature leave behind? [51, 52] | No scheduled cumulative consolidation before the checkpoint request; restructuring planned as backlog items |
| Transmission | Which implementations, rules and lessons shape the next change? [69, 83] | Growing specifications and read path; previously working identity setup not preserved |
| Selection and correction | Do review and tests expose the cumulative consequences before more work depends on them? [25, 92] | Proxy capture; process-gate inflation; feedback starvation |

The five categories in Table 4 are post-hoc interpretations. The supplement states their operational boundaries and the checks needed to distinguish their effects.

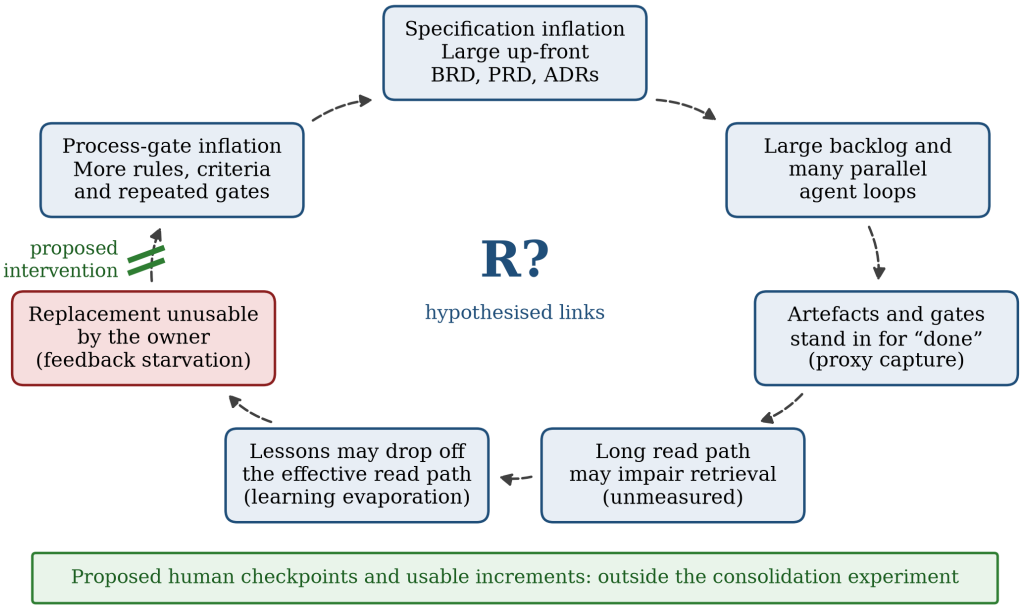



**Figure 4:** Proposed reinforcing relationships drawn from the case interpretation. Their causal effects remain unmeasured. Context overload is unmeasured. The proposed human-checkpoint intervention differs from the consolidation package tested in Section 7.

### 5.1 A proposed reinforcing relationship

The mechanisms suggest a possible reinforcing loop (Figure 4). In this case a large specification accompanied a large backlog and parallel sessions. Artefact-based gates generated more documents; some recorded lessons were absent from later briefs; regressions were not exposed through routine product-use feedback. The author responded to gaps with more rules, criteria and gates. That sequence is consistent with reinforcement, but does not establish each causal link. A running increment and a human checkpoint are proposed conditions of every cycle, to test whether they interrupt it.

### 5.2 Case validity and rival explanations

The five categories are not independent causes identified by these data. The same missing identity and staff bindings can indicate deferred migration work, incomplete deployment acceptance or a lost lesson. Discriminating them requires the deployment comparison and actual session-input trace identified in Table 1, not another count of documents or citations. Similarly, a growing read path does not demonstrate context overflow, and an expensive gate is not necessarily a disproportionate gate. Per-change gate time, avoided defects and human effort were not measured.

The source-layout migration, owner-selected priorities and delegated merging changed alongside coordination. Scope additions, budget constraints, model/tool changes and deployment conditions also differ between periods. These are direct rivals to an autonomy-only account. The supplementary rival-explanation register retains the full register, separating recorded circumstances from plausible but untested alternatives. Bird et al.'s study of co-installed applications on Windows machines supplies a methodological analogy for checking environmental influences on reliability [17]. It neither tests AI-agent configuration nor diagnoses this case's sign-in failure.

Construct validity is limited by whitespace counts, lexical indicators and selectively reviewed records. Appropriate caution can raise boundary-phrase density; missing references do not prove missing knowledge. As coordinator designer, participant and analyst, the author selected the

episodes and may favour a coherent retrospective explanation. AI synthesis is not an independent observer. Public records support recomputation but not independent corroboration of every statement. The single practitioner-designed exercise and purposively selected external examples do not establish prevalence or general causal effects. The proposed experiment has separate identification limits in Section 7.3.

# 6 Consolidation hypothesis and harness

The observed operational failure and the prospective hypothesis are distinct. Missing deployment identity and staff bindings are concrete obstacles to sign-in; structural code consolidation would not necessarily restore either. Recovery of that journey requires configuration and end-to-end verification. Consolidation is worth testing for a different reason: repeated feature work may make later changes harder, even when deployment dependencies are correct. The experiment probes that possibility through retained tests and CR1; it does not claim that consolidation would have prevented this episode.

The idea is already practised. OpenAI describes replacing weekly manual clean-ups with recurring background agents that open small refactoring pull requests, alongside a documentation-gardening agent [54]. Anthropic asks each session to leave a clean state and update a progress file [91]. Neither account reports a controlled comparison of feature-only delivery with consolidation between feature steps. Darwin's continuous optimisation is a further precedent for automated improvement, but its performance evaluations do not isolate that intervention or test an unforeseen change request [72] (Section 2.3). RefactorBench already evaluates refactoring agents [35]; the narrower hypothesis concerns retained behaviour and later adaptation during repeated feature additions.

AutoSaddler's preprint abstract already treats harness improvement as offline learning from execution traces, combining failure diagnosis, structured patches and validation-based update selection [70]. Agent-led reflective correction is therefore not the novelty claimed here. The proposed intervention instead consolidates the retained repository between feature steps and measures retained contracts and a withheld change request.

The proposed consolidation harness tests a bounded part of reflective correction. The distinction is what persists afterward. Self-Refine revises an output and Reflexion retains lessons for later trials [57, 81]; the present intervention instead consolidates the software and its handover between different feature requests. CRITIC's tool feedback is another precedent [38], but it does not authorize feeding the held-out evaluation into this experiment. Whether the retained changes help an unforeseen request remains the empirical question. The harness schedules each pause; the agent is not authorised to reject features or redirect the product. After every feature the loop stops, and an agent:

1. runs the tests and a metrics script that lists complexity hotspots, duplicated blocks and lint warnings;
2. reviews the last feature's diff and the hotspots against a short checklist (duplication, long or complex functions, unclear names, dead code, rules from one feature leaking into another, tests that check implementation rather than behaviour);
3. refactors in small, behaviour-preserving steps, each applied through `test && commit || revert` [9], so that every change either keeps all tests green or disappears;
4. may add tests that pin current behaviour, but may not change or delete the example tests that define the features;
5. rewrites a design note of at most 300 words that says where each rule lives and what debt remains, replacing the previous one rather than appending to it;
6. stops when a quality budget is met or its time budget runs out.

Only then may the next feature start. The supplement reproduces the historical v4 prompt; the current researcher kit defines the amended prompt. The intervention combines refactoring, additional tests and a design-oriented note; it is not a pure refactoring treatment.

The hypothesis draws on Beck's alternation of features and futures [12, 15] and *Tidy First?* [10]. It is not free, and may not work. Agent refactorings can be incomplete or incorrect [35]; human studies also caution against equating a refactoring label with a quality gain. Kim et al. report developers' refactoring challenges and benefits [47]; Bavota et al. find that observed refactorings do not consistently target or remove code smells [4]. These findings motivate checking outcomes rather than assuming improvement. Test-commit-or-revert protects only behaviour represented by its tests, not every intended contract [56]. Section 7 specifies how the package will be evaluated.

# 7 Testing the harness: a prospective experiment

The experiment asks whether scheduled correction improves the repository inherited by later feature work. RQ3 is addressed by the prospective design specified below. The 1 November amendment supersedes v4's two runs per arm/model. Historical freezes, diagnostics and the original owner-reported OSF registration are retained; none approves this new design. The researcher kit requires a new source-bound freeze and prospective OSF amendment before collection. Agent configuration protocols and empirical-study guidance inform this pre-data design [3, 55, 89].

## 7.1 Design and estimands

One independent replicate is a fresh repository trajectory through S1's ten ordered features and a final held-out change request, CR1. Every selected model, GPT 6.1 Sol, Sonnet 5.5 and Gemini 4 Argon, runs every arm. Randomised interleaving within model/replicate blocks uses seed 2027. Feature steps, tests and checkpoints are repeated observations within a trajectory, not independent replicates.

**Table 6:** Prospective five-arm allocation. D is the matched-resource A+ control; E tests visible regression repair, not hidden-test feedback.

| Arm | Intervention and information | Per-cycle ceilings |
|---|---|---|
| A | Feature delivery only | 30 min; 300 nominal credits |
| B | Feature, then behaviour-preserving consolidation, additional tests and replacement DESIGN.md | 30 + 15 min; 300 + 300 credits |
| C | Feature, then read-only review and replacement REVIEW.md | 30 + 15 min; 300 + 300 credits |
| D (A+) | Longer feature delivery, without a scheduled consolidation pause | 45 min; 600 credits |
| E | Feature, then repair of previously specified behaviour using visible specifications and examples | 30 + 15 min; 300 + 300 credits |

B–D is the primary practical comparison: consolidation versus giving the builder the same total time and nominal credits. Equal ceilings do not guarantee equal realised resources; D also has fewer fresh sessions. B–A measures the additional package; B–C also changes editing rights, tests and handover content. E–B contrasts visible behavioural repair with behaviour-preserving consolidation. These secondary package comparisons do not isolate refactoring or any of the five case mechanisms.

C cannot repair code during its review session, but REVIEW.md can direct repairs in later feature sessions. Follow-through needs review-item and diff locators with external adjudication; file overlap alone is not proof. B's added tests must agree with visible specifications, rather than legitimating a defect. Subsequent feature sessions must keep accepted tests green. E can correct an agent-authored expectation only with a specific contradictory specification and a retained explanation. Accepted Python test files are immutable in full, including imports, fixtures and module/class constants; additions use new files. E retains only its explicit specification-bound correction exception. Immutable example tests remain protected in every arm.

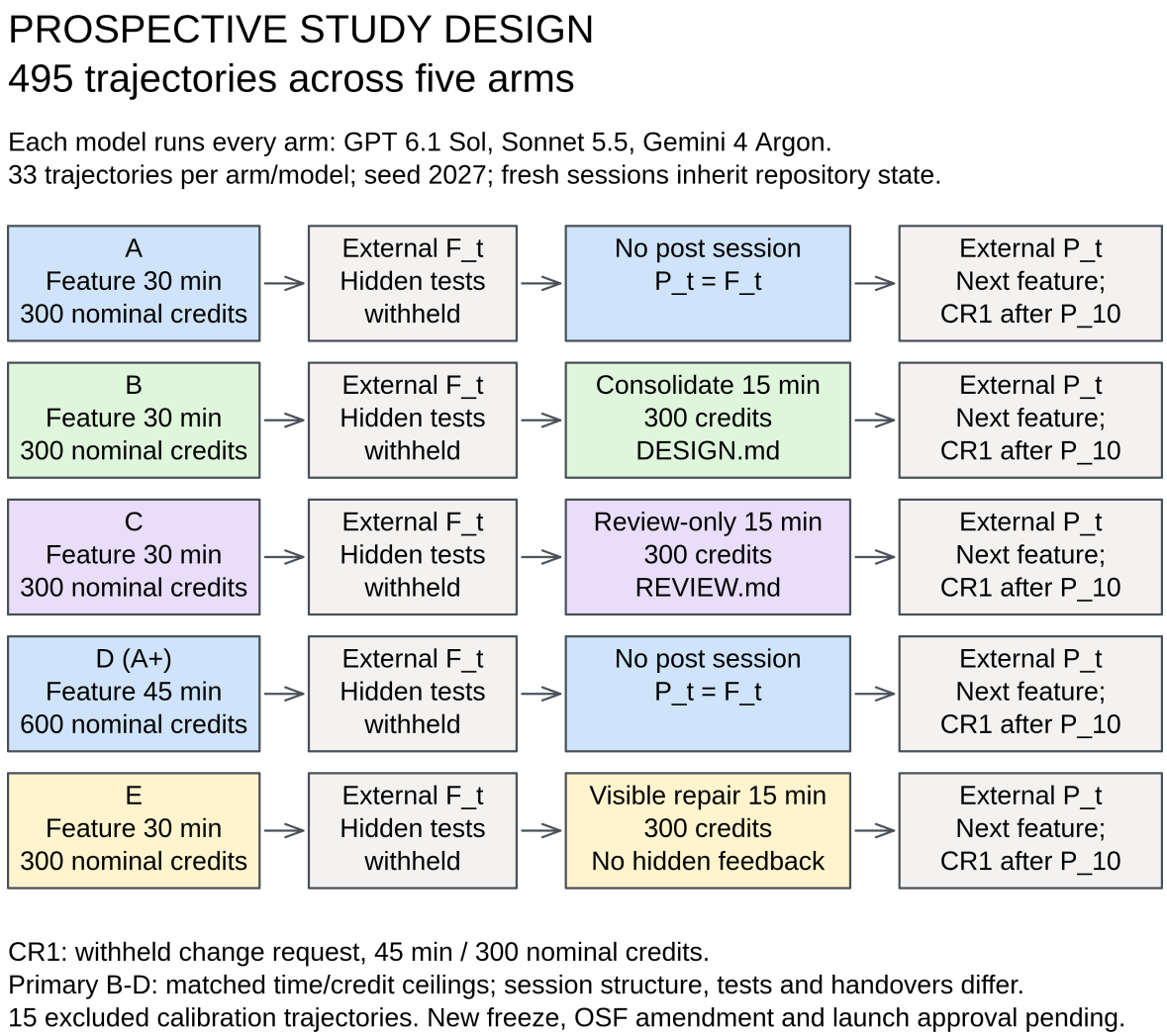



**Figure 5:** Prospective design, not results. Every model runs all five arms. The external evaluator is separate from candidate tools. CR1 content remains withheld until its session; privacy and isolation reduce exposure without proving that a model has no prior knowledge.

### 7.2 Outcomes and retained-contract loss

The primary outcome is final behavioural coverage: contracts passed out of all 127 scheduled feature contracts at $P_{10}$, before CR1. Coverage combines acquisition, retention and repair. Retained-contract loss and CR1 adaptation are distinct secondary outcomes, with legacy identifiers H2 and H6 retained only for data traceability. HIR/IDR definitions are in the supplementary appendices; comprehension is not measured.

For trajectory $r$, $F_t$ is the retained state after feature step $t$ and $P_t$ the state after its assigned

post-step; these source states coincide in A and D. Let $\mathcal{R}_{r,t}$ contain eligible, unchanged contracts that passed at least one earlier end-of-cycle checkpoint $P_u$, $u < t$. Superseded contracts leave this risk set. At $F_t$ and $P_t$, let $\mathcal{O}^p_{r,t} \subseteq \mathcal{R}_{r,t}$ contain contracts with observed binary results $S^p_{r,t}(q)$, $p \in \{F, P\}$. Report

$$L^p_{r,t} = \sum_{q \in \mathcal{O}^p_{r,t}} \mathbf{1}\{S^p_{r,t}(q) = 0\}, \qquad d^p_{r,t} = \frac{L^p_{r,t}}{|\mathcal{O}^p_{r,t}|}. \tag{1}$$

The density is checkpoint prevalence of failure among observed, previously acquired contracts. An empty observed denominator is undefined. Publish the full risk set, observed subset and missing count $M^p_{r,t} = |\mathcal{R}_{r,t}| - |\mathcal{O}^p_{r,t}|$ alongside it. The full-risk loss count lies between $L^p_{r,t}$ and $L^p_{r,t} + M^p_{r,t}$; missing evaluations cannot be treated as passes. Never-acquired contracts remain acquisition failures. Retain onset, recovery, recurrence and uncertainty across missing checkpoints. Recovery does not erase an episode. Distinguish same-cycle post-step losses and rejected attempts from losses in retained code.

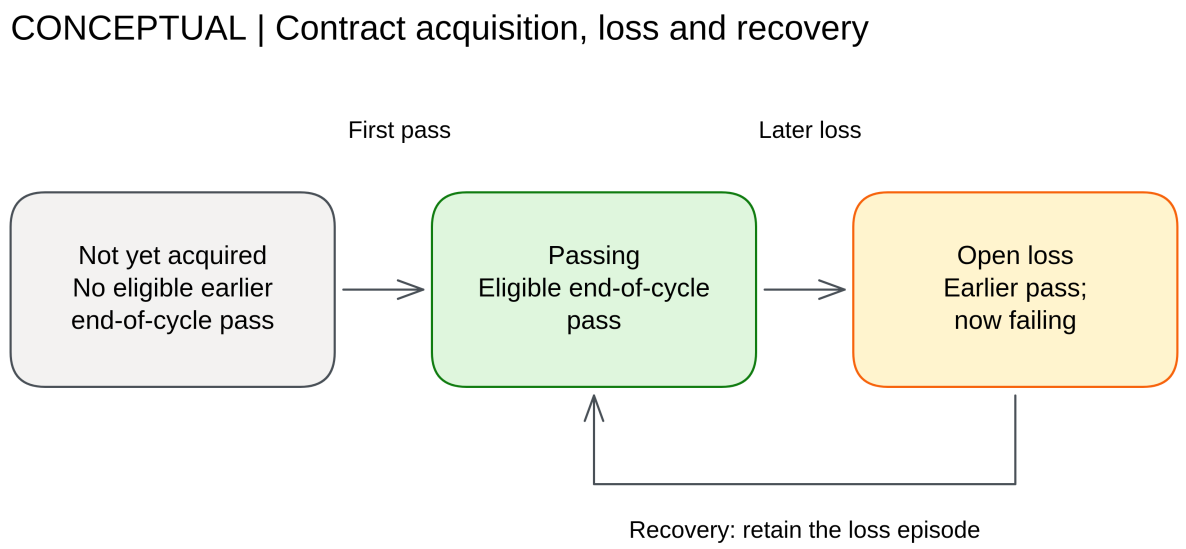



**Figure 6:** Conceptual contract states. Acquisition, loss, recovery and unobserved evaluation are distinct; the diagram contains no measured frequencies.

CR1 success requires all 15 CR1 contracts and retention of every feature contract passing before CR1. Also report contracts passed out of 15, previously passing contracts lost, and observed/missing denominators. Stratify by whether F9 and F10 prerequisites were acquired before CR1: an incomplete arm has fewer retention obligations, and prerequisite strata are post-treatment selections, not randomised subgroups. One fixed CR1 is task-specific. A prewritten, randomly selected request pool belongs to a separately registered subsequent study.

Actual AI credits include every delivery and CR1 attempt, including failures, assistance and infrastructure retries; phases are reported separately. Cheap failure is not efficiency. Credits, tokens, elapsed time and currency are distinct. Lizard measures complexity, Ruff lint and jscpd duplication. External fixed settings and `ruff -isolated -ignore-noqa` prevent local suppressions from silently improving the lint score; count added suppressions separately. SDK request wall time and tool time are observable. Provider-active time and rate-limit waits require explicit telemetry; absent telemetry is not zero.

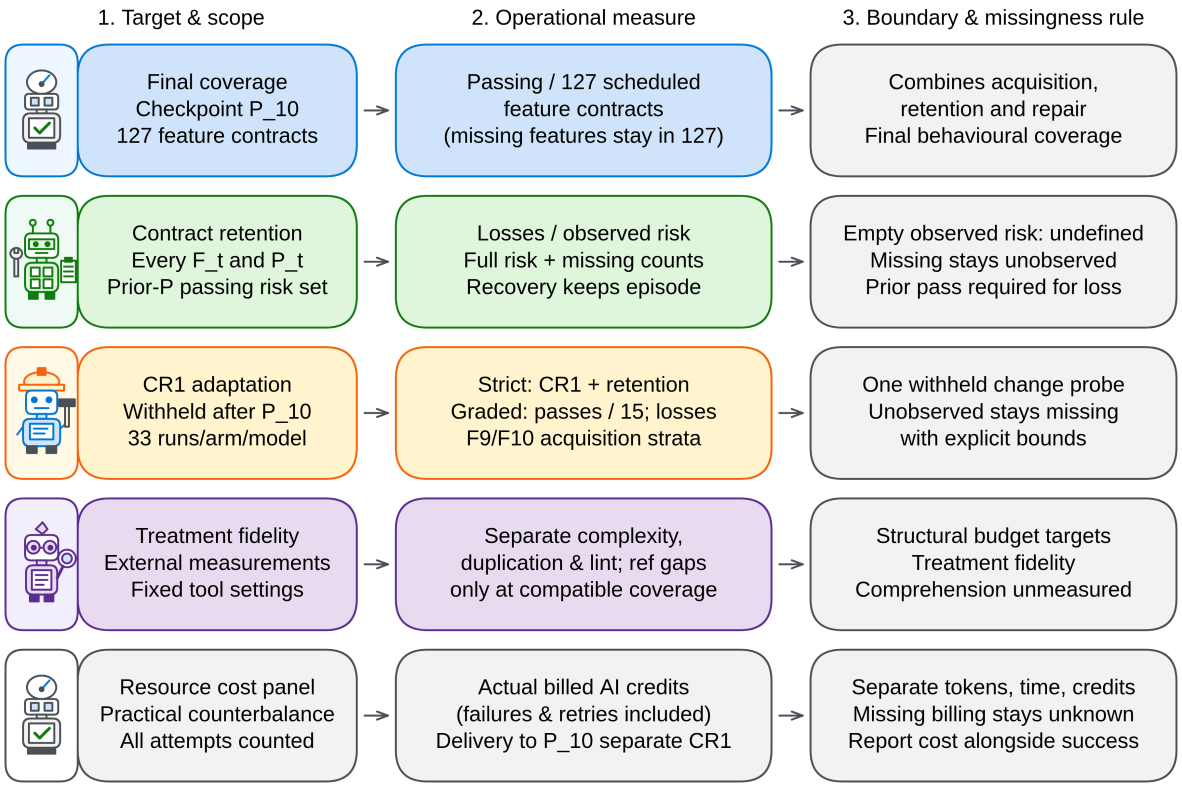



**Figure 7:** Coverage, retained behaviour, adaptation and all-attempt cost are separate measures. Structural metrics targeted by B's stopping budget are treatment-fidelity checks, not independent efficacy outcomes. Unknown results and billing remain unknown.

## 7.3 Replication and uncertainty

The author selected a minimum useful absolute coverage gain of ten percentage points, approximately thirteen test-weighted contracts. Coverage scores $Y$ are expressed on the 0–100 scale. Let $\overline{Y}_{ma}$ be mean final coverage for model $m$ and arm $a$. The primary estimand is the equal-model-weight difference in mean coverage over the $M = 3$ selected models. Its estimator is

$$\widehat{\Delta}_{BD} = \frac{1}{M} \sum_{m=1}^{M} (\overline{Y}_{mB} - \overline{Y}_{mD}). \tag{2}$$

For balanced cells of $n_{\text{cell}}$ independent trajectories and a common within-cell variance $\sigma^2$, its planning variance is $2\sigma^2/(Mn_{\text{cell}})$. The assumed SD is 25 percentage points. For one two-sided comparison with $\alpha = 0.05$, target power $1 - \beta = 0.80$ and target difference $\Delta_* = 10$ points, the normal planning approximation is

$$n_{\text{cell}} = \left\lceil \frac{2(z_{1-\alpha/2} + z_{1-\beta})^2 \sigma^2}{M\Delta_*^2} \right\rceil = 33. \tag{3}$$

This gives 99 trajectories per arm, 495 overall. The target power is conditional on independent trajectories, the assumed variance and the blocked estimand; it is not guaranteed. SD assumptions of 15, 20 and 50 points require 12, 21 and 131 per cell respectively. The bounded score's worst-case SD is 50; the 25-point assumption needs explicit author and prospective registration approval.

Resample whole trajectories within model and arm, with 10,000 draws and pointwise 95% percentile intervals; disclose missing frames rather than silently pool observed models or complete cases. Model-specific estimates and other contrasts are secondary, with no family-wide coverage claim. Retain scheduling-block identifiers. Before collection, source-bound review must reconcile this independent-trajectory variance and bootstrap scheme with the model/replicate blocking: residual dependence would require a block-preserving analysis and revised planning variance. Chen, Zaharia and Zou's dated ChatGPT comparison motivates monitoring service drift despite exposed names [22]; it does not measure drift in this roster or make a version pin an immutability guarantee. Report the pooled within-cell SD with its degrees of freedom for future planning. For binary CR1, report per-model success counts and exact binomial component intervals under independent runs with a common success probability within each cell. Aggregate contrasts must preserve model strata. The coverage calculation supplies no power guarantee for CR1.

### 7.4 Resources, feasibility and collection gates

The unchanged envelope is 300,000 nominal credits, with 270,000 local admission, prior spending and unknown historical provisions retained. Original 300-credit sessions use a 270-credit soft limit; D's 600-credit feature uses 540. A retry shares the original allowance. In-flight responses can overshoot; these are not provider billing caps. No verified dollar conversion is available.

**Table 7:** Planned resources, not expenditure. Worst-case allowances exceed the aggregate envelope; affordability is a real launch gate.

| Arm | Trajectories | Sessions | Maximum agent-hours |
|---|---|---|---|
| A | 99 | 1,089 | 569.25 |
| B | 99 | 2,079 | 816.75 |
| C | 99 | 2,079 | 816.75 |
| D | 99 | 1,089 | 816.75 |
| E | 99 | 2,079 | 816.75 |
| Study total | 495 | 8,415 | 3,836.25 |
| Excluded calibration, no CR1 | 15 | 240 | 105.00 |

At nominal caps the combined allowance is 2,902,500 credits, not a permitted expenditure. Cost-only excluded calibration must show that the entire fixed allocation fits remaining admission and the deadline; otherwise halt and amend before collection, never quietly reduce the sample. Proposed progression criteria are infrastructure failures at most 5%, no missing primary observations, complete billing/model/help records, zero oracle contradictions in triplicate reference checks, and no information-boundary failure. Hidden-test mutation assessment must report its actual mutant denominator and unresolved equivalence. These are feasibility criteria, inspired by pilot-study guidance [30], not a clinical CONSORT compliance claim.

Only current and previous visible specifications and examples reach candidate tools. Hidden tests, scores, identifiers, messages, reference code and future CR1 remain external. Candidate commands use no-network containers; the trusted host SDK's provider connection is different. The independent researcher receives a protected kit without access to the original repositories or manuscript. Use a researcher-managed workspace/account without the author's original checkouts, manuscript or session-context caches; record actual access and prior exposure rather than assuming blindness from a private flag. The owner will make both repositories private before collection and release approved materials with the paper afterward. OSF registration/embargo and GitHub privacy are separate; neither removes prior public or model exposure. Fresh source-bound access checks, approval, freeze, prospective OSF recording and launch permission remain required.

Unsuccessful feature increments are reset; later features and scheduled B/C/E post-steps continue on retained green state. Hidden denominators still include undelivered features. One infrastructure-only retry shares the original budget; poor performance never justifies replay. Contradictory valid oracle results halt collection. Assisted and incomplete trajectories and all their costs are retained, without replacements or success-conditioned primary analyses.

# 8 Discussion

## 8.1 Projected cost of omitting consolidation

The experiment measures costs over ten features and one final change. It does not yet estimate the price of maintaining a larger system without refactoring. Figure 8 and Table 8 therefore give an explanatory sensitivity scenario, not results, an empirical forecast or an addition to the frozen analysis. The economic hypothesis is that early investment can reduce the implementation and repair burden inherited by later features. It can fail: refactoring may consume credits without reducing that burden or may damage behaviour. Cost must always be read beside coverage, retention and adaptation.

Let $N \geq 1$ be an integer number of feature cycles, $k$ the assumed average consolidation investment in AI credits per cycle, and $\delta \cdot (t-1)$ the assumed excess feature-implementation cost in A relative to B at cycle $t$, excluding B's consolidation session. Shared baseline implementation costs cancel. Under the explicit simplifying assumptions of constant $k$, a linearly growing relative premium $\delta \geq 0$, and comparable delivered behaviour, the cumulative net difference is

$$\Delta C(N) = C_A(N) - C_B(N) = \sum_{t=1}^{N} [\delta(t-1) - k] = \delta \frac{N(N-1)}{2} - kN. \tag{4}$$

The first term is the assumed gross avoidable burden of omitting consolidation; the second is the investment to avoid it. A positive difference means A costs more under these assumptions, not measured savings. For $k > 0$ and $\delta > 0$, the first cycle count with $\Delta C(N) \geq 0$ is $N_{\mathrm{BE}} = \lceil 1 + 2k/\delta \rceil$; when $\delta = 0$, the investment never pays back in this credit-only model. Real trajectories need not follow a linear premium, and B can retain debt too: $\delta$ is a relative gap, not A's absolute debt or a claim that B is debt-free.

**Table 8:** Illustrative net A–B cost in AI credits from Equation 4, with assumed $k = 5$ credits/cycle. Negative values mean B's investment has not paid back; zero and positive values mean break-even and a conditional credit advantage. None of the scenarios is an observed or preferred estimate.

| Assumed premium increase $\delta$ | 10 cycles | 20 cycles | 40 cycles |
|---|---|---|---|
| 0 credits per successive cycle | -50 | -100 | -200 |
| 0.5 credits per successive cycle | -27.5 | -5 | 190 |
| 1 credit per successive cycle | -5 | 90 | 580 |

The study can report actual B consolidation consumption and all-attempt CR1 cost by model. Calibration can anchor $k$ once billed spend exists. Because feature budgets are capped, harder work can instead reduce coverage: $\delta$ is an uncapped, quality-matched counterfactual, not a directly identified endpoint of this design. Even the enlarged replication over ten features does not identify a stable long-run $\delta$ or justify extrapolation to 40 features. A future forecast would require longer replicated sequences, matched delivered quality, measured residual repair work and independently checked billing. Human review, operational failures, opportunity costs and currency prices are not measured by this kit and cannot be monetized from its credits or code-size proxies. Thus the current projected cost of inaction is conditional and unestimated, not a claimed business saving. For comparison, assumed $k = 100$ and $\delta = 1$ yield break-even at 201 cycles, not eleven. No observed consolidation spend supports a preferred $k$ yet.

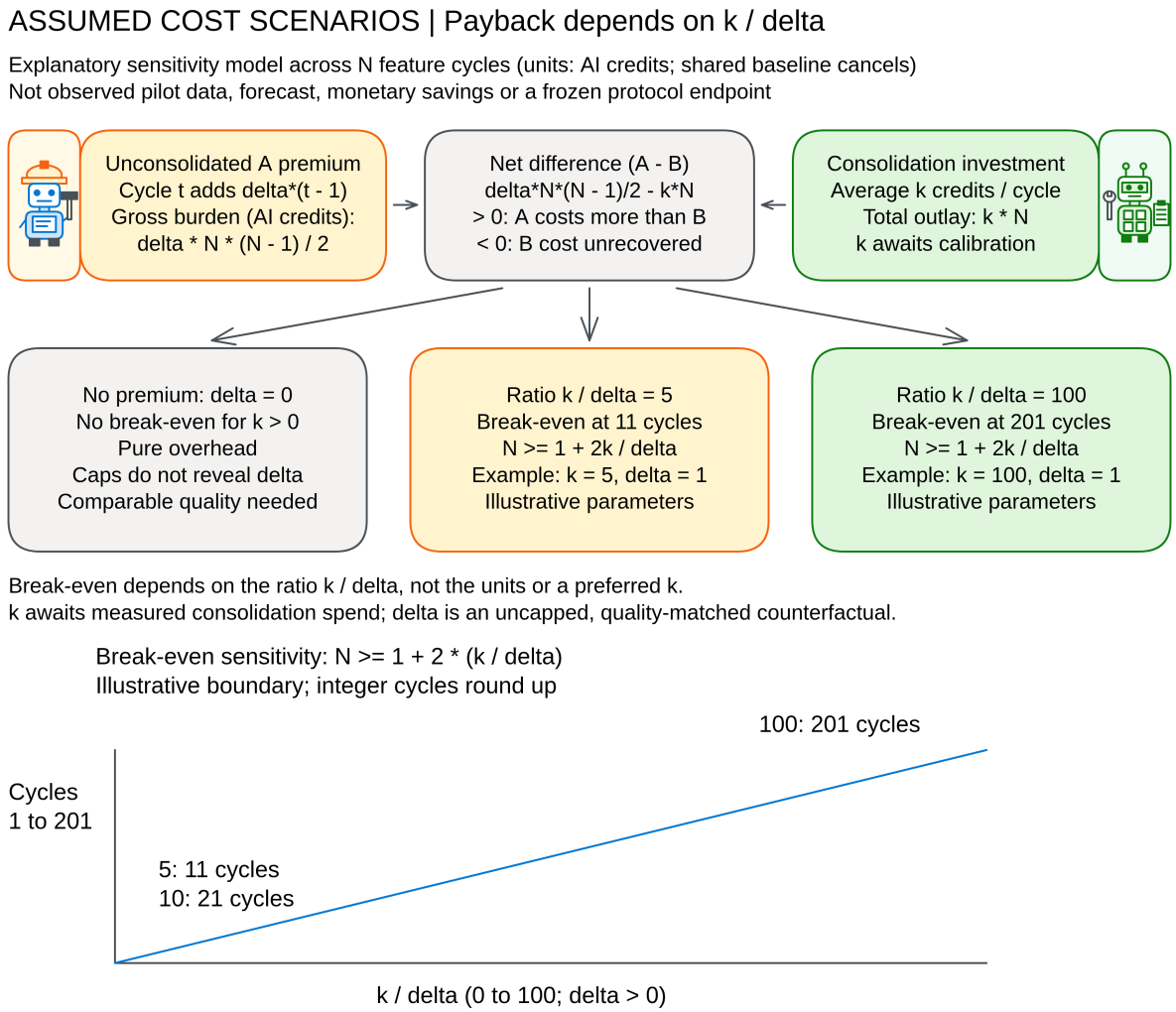



**Figure 8:** Assumed cost-of-inaction mechanism and break-even sensitivity, not pilot results or an empirical projection. A linearly increasing relative implementation premium accumulates across feature boundaries; recurring consolidation has an explicit cost. Payback depends on $k/\delta$: a ratio of 5 breaks even at 11 cycles and a ratio of 100 at 201; $\delta = 0$ gives no payback for $k > 0$. Parameters are invented for transparent sensitivity, not fitted to study data. No dollar conversion or human-work estimate is implied.

## 8.2 What worked

Agents contributed useful challenge and rapid iteration. The research phase assembled a corpus and analysis in days; a human-only duration was not measured. The role agents' challenge killed a bad idea, five end-to-end agents, before any code was written. The agents refused to invent approvals, figures or company facts, and recorded the limits of their evidence. The MVP was built and reshaped in a day. These episodes included targeted human challenge and feedback; they are not a controlled comparison with human-only teams or proof that all short loops succeed.

## 8.3 Human judgement and reflective work

The participant supplied several kinds of reflective work: choosing what not to build, recognising the unusable replacement and deciding which earlier lessons mattered. Naur's theory-building account helps describe that retained understanding [65]. Beck's distinction between visible output and understanding supplies another framing [14]. These judgements were supplied by the

author during the case. Whether agents could make comparable judgements remains open. The narrower experimental question is whether bounded post-feature correction preserves behaviour and later changeability.

Human sign-off also need not establish who formed a judgment. In Anthropic's randomised study, 61 analysts were recruited and 50 used the agent. Those 50 analysts matched their agent's randomly assigned preference on 86% of four analytical choices [39, Study design; Results]. The preferences were deliberately strengthened in one farming-analysis task with one model; the rate concerns a different task and setting from coding [39, Our study's limitations]. This challenges an assumption that retaining approval alone preserves independent judgment, not the case's reported decisions. The article proposes asking experts for their view first or making the best case for each option [39, What this means]. Separately, recording alternatives and a human's reasons before accepting an agent's recommendation is the present author's proposed safeguard, not a remedy evaluated in that study.

### 8.4 Can another agent restore futures?

Beck's alternation suggests preserving the options consumed by feature execution [12, 15]. Darwin, AutoSaddler and SkillOpt already provide precedents for agent-guided correction; reflection itself is not the proposed novelty. Ng's reflection framework also argues for evaluating and revising work rather than treating first-pass generation as the finished product [66]. Improvement here means a measured advantage in retained behaviour or later adaptation, not merely writing a lesson, changing a prompt or calling the process self-improving. The experiment does not train model weights; the evolving repository and handover are the states under test. An exposed model name is not a guarantee against provider-side changes.

The memory presentation's proposed "dreaming" pass is a useful contrast: it curates prior transcripts into persistent context [5], whereas this study changes retained code, tests and the handover. Memory updates can carry an incorrect lesson forward just as code updates can retain a defect. A self-improvement label therefore cannot replace the retention and adaptation measures. The study asks what remains after a fixed repository-consolidation step, with CR1 as one future-change probe. It does not test choosing when to interrupt delivery, changing product goals or restoring a deployment. Those questions, and matched human/agent steering, belong to the separate research agenda in the ancillary supplement.

### 8.5 Implications for practice

The case motivates protecting a usable increment, bounding the delivery horizon and keeping lessons retrievable or executable. The ancillary supplement retains the proposed practice checklist and framework-builder suggestions. For future product integrations, routine intermediate tracking files should stay out of the squash-merged product diff once durable findings and original review evidence have been preserved. This is a proposed practice, motivated by the large intermediate evidence trails retained in this case. It is not permission to wipe an active or historical tracking tree: selection must be finite, archive-backed and restore-verified, with adverse reviews retained. Proportionality never overrides mandatory safety, privacy, approval or publication gates, and compact context views must retain their source evidence. The pilot does not test that practice package, an optimal human-intervention cadence or the preservation of expertise. Those remain separate questions.

## 9 Conclusion

The unit of progress in cumulative development is the system left for the next change. Knight Capital shows how a new deployment can activate an old fault. WyCash makes consolidation part of continued growth. Agent research adds a direct cross-feature test: in SWE-Maintain,

earlier changes affect later tasks even when conversation history is discarded. Its clean-history results make preserving behaviour a more precise target than treating all agent-written code as structurally deficient.

The instructions governing development need maintenance too. Repository studies show repeated additions and revisions, alongside pruning; controlled context-file comparisons show that more guidance can increase work without improving task resolution. Code, tests and instructions are all inherited state, but they require different measures of correctness and burden.

The motivating case traces a fuller process: agent-assisted research and challenge shaped requirements, prototypes and a hierarchical backlog; autonomous issue delivery then failed to preserve the participant's usable workflow. Renewed feedback, refinement and refactoring redirected subsequent work. The five candidate mechanisms frame possible links among accumulated obligations, lost lessons, completion proxies and insufficient product-use feedback. Migration and deployment dependencies remain central to that particular failure. The versioned repository retains the evolving record and the distinction between local engineering evidence and accepted journeys (Section 4.4).

The prospective five-arm study asks what scheduled consolidation leaves for later work. It separates final behavioural coverage from retained-contract loss, probes an unforeseen change and reports the resources consumed. Secondary metrics provide descriptive evidence; testing the proposed mechanisms would require further work. A new freeze, prospective OSF recording and collection gates precede the planned runs. The cost-of-inaction scenario makes the break-even assumptions explicit. What matters is whether the agents' combined work preserves the behaviour and options needed by subsequent features.

## Acknowledgements

The author thanks the colleagues and partners whose questions shaped this report.

## Use of generative AI

GitHub Copilot with HVE Core agents supported development of the system under study. The case method distinguishes reported interface labels from recorded selectors; the deployed assistant exercise used `gpt-4.1-mini-2025-04-14`, not the coding-agent model. Microsoft 365 Copilot supported summarisation and reference discovery.

GitHub Copilot in VS Code assisted log analysis, mechanism naming and organisation, manuscript drafting and revision, bibliographic checks, public repository inspection, figure preparation and local validation. Scientific Agent Skills supplied developmental writing and methodology guidance, including the SymPy, statistical-analysis, scientific-critical-thinking and experimental-design skills for this equation review [45]. This citation records assistance tooling, not evidence for the case findings or consolidation hypothesis. AI assistance supplied neither an independent empirical coder nor replicated application or study results. The package README retains the revision-by-revision account and source-access limits. GitHub Copilot CLI assisted the 29 October source integration and scoped method-clarity revision. For the subsequent editorial and visual revision, Gemini 4 Argon proposed infographic concepts. Those suggestions were checked against the manuscript and adapted into editable Excalidraw conceptual diagrams; they are not experimental outputs or independent validation. The earlier 1 November v8 revision reconciled the experiment description with v4, added the measurement and conditional cost-of-inaction explanation, and used Gemini 4 Argon for editable infographic design and implementation. AI-generated conceptual graphics and assumed economics are explicitly separate from experimental data; human author approval remains required. The later v9 revision used Copilot-assisted source comparison, static measurements, five-arm kit implementation and local

native diagram authoring. It added no experimental trajectories or independent human peer review, and made no new Gemini design-session claim.

The author supplied the observations, selected the case and made the analytical judgements, and reports checking the repository figures and Azure expenditure and confirming the Beck paraphrase after reading the essay and watching the talk. The supplied automatic critique and this revision's focused source and arithmetic checks are assisted evidence assessments, not independent human verification. The package records their coverage and unavailable sources. Metadata checks do not establish claim support; source comparison does not independently validate a source's experiments. The author takes responsibility for the content. Further disclosure must follow the actual submission venue's policy, which has not been assessed here.

## Competing interests

The author is a Microsoft employee studying Microsoft and GitHub tools within a Microsoft training programme and designed the coordinator under study. This professional and participant-observer interest is disclosed; it does not establish an independent evaluation or company endorsement. External-publication clearance remains a separate author responsibility.

## Data availability

The repository analysed was publicly accessible during the case at https://github.com/jmservera/ama; this report uses commit `29c6b512f5e68733bb2cc1882343db39f5eb2cad`. The separate 29 October repository follow-up pins `9b74ac8daaec2a880172e43161b37ac7f2c3f801`; its selected API records, retrieval interval, merge-window definitions, document hashes and metrics are in anc/repository-metrics-2026-10-29.json. The log-measurement and figure-generation scripts are provided with the source. The ancillary file anc/case-evidence-matrix.json indexes O1–O5, including reported locators and missing release/execution links; it is not newly collected corroboration. The companion anc/supplementary-context.md retains practitioner material, external observations, proposed practices and deferred extensions. The ancillary file anc/repository-observations-2026-10-25.json records the additional repositories' source revisions, tracker counts, sampling boundary and claim locators without reporter identities or attachments. Live issue states can change independently of the pinned source commits. The ancillary experiment notes in HVE Core's Minimum Viable Experiment format are earlier proposals, not the tagged study record. The separate researcher kit is at https://github.com/jmservera/ama-thought-mve; use its historical frozen v4 release. The revised design requires a new exact-source packet and prospective OSF amendment before collection; v4 does not approve it. The author plans to make both repositories private before the experiment, give the independent researcher a protected kit without access to the original repositories or manuscript, and release the approved kit and experiment materials with the paper after collection. OSF registration and embargo are distinct from GitHub visibility. Prior public exposure cannot be undone, and privacy alone cannot prove absence of model contamination. On 1 November the owner explicitly authorised development PR updates while both repositories were public. That interim exposure is separate from the planned post-collection release and must be disclosed to the researcher. Private paid diagnostic code, results and transcripts and runtime image bytes are excluded from that packet. The author-reported original OSF registration does not establish prospective v4 registration. Sanitized pilot data and exact execution provenance still require a later authorized deposit; no completed independent replication is claimed.

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# A Supplement

This section supplies source-specific qualifications, reproducibility detail and unperformed discriminating checks beyond the main account. It does not repeat the Background, case chronology or figures. The full current protocol is in Section 7; the details below retain their individual source and measurement boundaries.

## A.1 Additional source qualifications

The details below qualify specific uses of the literature in Section 2; they are not additional results from this study.

**Table 9:** Source-specific qualifications relevant to the proposed measurements.

| Source | Additional detail | Consequence for interpretation |
|---|---|---|
| Sinha et al. [83, Sec. 3.2, Figs. 5–6] | The artificial-history experiment varies erroneous prior updates. Qwen3 with thinking enabled avoids the reported effect in that setup, whereas scaling non-thinking models does not remove it. | Earlier-error sensitivity depends on configuration; it is not a universal property of all models or a measured repository-decay rate. |
| Pan et al. [69, Sec. 5.1, Fig. 4] | In the `parse_crontab` episode, a local Python 3 change satisfies current tests but omits module-wide compatibility aliases later code uses. | An initially successful patch can leave a latent dependency failure. This complements, rather than repeats, the line-terminator example in the main text. |
| Gloaguen et al. [36, Secs. 4.2–4.4] | File length does not explain success in their analysis, and generated context helps when other documentation is removed. | Instruction length alone is not an adequate exposure or burden measure; available alternative documentation matters. |
| Jiang et al. [44, Secs. 5.2–5.5] | Skill misapplication and missing runtime verification remain failure modes. The retrieval diagnostic is separate from downstream execution. | Lower exact-match access alongside sustained task success does not isolate an effect of retrieval; accessed, applied and sufficient guidance differ. |

The provenance of merge-outcome studies also matters. Peralta et al. start with 9,799 human-reviewed agent PRs and manually inspect 717. Their 35.7% clear-failure share among rejections and 15.4% explicit reviewer-involvement share among merges describe that manual sample [73]. Murphy-Hill et al.'s roughly 24% higher merge count among adopters is an observational rollout estimate, not delivered value [64]. AIDev's 932,791 agent-authored PRs from five agents across 116,211 repositories provide a population for research, not a count of accepted user outcomes [53].

Pavlenko et al.'s World Model offers an additional context-management design: preserve raw evidence, derive rebuildable views and retrieve compact, task-specific context with provenance and access controls [72]. That architecture is an alternative to relying on an expanding read path; its effect on the AMA case is untested.

## A.2 Case rival-explanation register

The main interpretation is in Section 5.2. The register preserves recorded rivals separately from untested alternatives. A limit is not grounds for ruling an explanation out. Score/deployment correspondence remains unverified (Section 3.6).

**Table 10:** Rival explanations with direct case records.

| Explanation | Case support | Counter-evidence or limits | Treatment |
|---|---|---|---|
| Migration decision | ADR 0006 (issue #196) froze feature work in the MVP and prioritised a new layout. | Health and restart checks passed, not usability. | Migration and coordination changed together; autonomy alone is not identified. |
| Capstone rubric | Research and documentation are rewarded; full completeness marks coexist with blocked sign-in. | A working demonstrator is required; release/scope linkage is pending. | Possible incentive, not a measured effect. |
| Owner overrides and priorities | Committed RPI records, full gates and delegated merging; PR #445 required an exception. | The MVP used agents successfully; HVE Core defaults retain human stops. | Owner choices are part of the case. |
| Model and tool changes | Versions changed during the study. | No matched rerun under fixed versions. | Actual session configuration remains partly unknown. |
| CI and cloud budgets | Issue #413 moved validation locally; costs prompted a higher ceiling and reduced resources. | Merges continued without established deployed usability. | More budget is not proven to resolve delivery. |
| Identity and sign-in setup | Missing registration/staff bindings; issue #349 distinguishes redirect from sign-in. | Those steps worked in the MVP. | Concrete configuration obstacle, not proof of code decay or lost knowledge. |

**Table 11:** Additional plausible rivals and unperformed discriminating checks.

| Explanation | Reason considered | Counter-evidence or limits | Evidence still needed |
|---|---|---|---|
| External services and deployment drift | Model/identity configuration differs from the MVP. | No provider outage is established; health checks were narrower. | Versioned deployment/configuration and dependency-health records for the journey. |
| Scope expansion | RAI and consumption criteria added to issues; further resident-preview obligations. | Expansion predates autonomy; some controls are justified. | Original/added scope, sequencing and time-to-outcome comparison. |
| Inadequate acceptance tests | Checks passed without completed real-user sign-in; scorer read artefacts. | Bounded, honest tests need not be poor tests. | Capability-to-task-oracle mapping and execution evidence. |
| Rate and context limits | Parallel sessions and a large read path. | No runtime rate-limit, truncation or attention dataset analysed. | Actual inputs and blocked/retried request events. |
| Specification ambiguity | Selective review of long, cross-referenced requirements. | Important domain/consent decisions survived; ambiguity not independently coded. | Independent interpretation and conflicting acceptance examples. |
| Ordinary prototype debt | Rapid MVP followed by restructuring and rebuilding. | MVP worked intermittently; no comparable human-led migration. | Failure analysis distinguishing inherited debt from introduced faults. |

## A.3 Candidate-mechanism evidence audits

These are operational boundaries for this paper, not validated necessary and sufficient conditions for all agentic development. The indicators are already reported in Table 4, Table 1 and the case chronology; they are not repeated here. Experimental implications state what evidence would

discriminate an explanation; the tagged experiment does not manipulate the five candidates separately.

### A.3.1 Specification inflation

| Element | Audit |
|---|---|
| Definition | Requirements and acceptance obligations expand faster than independently validated behaviour, without evidence that the added scope is needed for the next usable increment. Length alone is insufficient. |
| Counter-evidence | Growth began before autonomy and sometimes followed the owner's feedback; regulatory and consent controls can justify detail. |
| Rival explanations | Genuine scope expansion, specification ambiguity, capstone incentives and an ambitious migration rather than agent-driven inflation. |
| Experimental implication | Compare smaller requirements horizons at matched scope and resources. The current fixed-suite experiment cannot identify this mechanism, even if consolidation helps. |

### A.3.2 Proxy capture

| Element | Audit |
|---|---|
| Definition | A completion proxy becomes a decision criterion and remains positive when an independent test of the intended outcome is negative. Proxy use alone does not establish capture. |
| Counter-evidence | Agents explicitly reported unmet acceptance; neither fabricated approval nor deliberate deception is established. |
| Rival explanations | An inadequate scorer or acceptance oracle, different evaluation targets, and a deployment failure invisible to repository review. |
| Experimental implication | Disagreement between visible completion and hidden behavioural tests would expose proxy limits. It would not by itself establish intentional test gaming or explain why a gate was chosen. |

### A.3.3 Learning evaporation

| Element | Audit |
|---|---|
| Definition | A recorded lesson relevant to a later decision is absent from, unusable on, or not applied through that decision's effective context path, and the associated problem recurs. A missing citation alone is insufficient. |
| Counter-evidence | Important domain decisions survived; the lesson could have been known but deferred. Actual session attention was not measured. |
| Rival explanations | Deliberate migration sequencing, deployment-specific configuration, context limits, or conflicting requirements rather than lost knowledge. |
| Experimental implication | Trace actual inputs and repeated faults. B–C cannot isolate memory quality because DESIGN and REVIEW notes and acting permissions differ. A matched-context study would be needed. |

### A.3.4 Process-gate inflation

| Element | Audit |
|---|---|
| Definition | Additional procedural obligations increase per-change burden without demonstrated proportional improvement in the targeted outcome. More gates alone are not evidence that their net benefit is negative. |
| Counter-evidence | Checks caught bounded defects and enforced evidence integrity. Gate duration and human review effort were not measured per PR. |
| Rival explanations | Legitimate safety and privacy requirements, budget constraints, new deployment complexity and owner-selected risk tolerance. |
| Experimental implication | Measure gate time and accepted outcomes under matched risk. The pilot's resource description concerns the package, not the effect of removing this repository's required gates. |

### A.3.5 Feedback starvation

| Element | Audit |
|---|---|
| Definition | Independent outcome-level feedback is insufficient to correct delivery decisions despite abundant artefact-level validation. The defining gap is between the evaluated proxy and intended outcome, not test count alone. |
| Counter-evidence | Human oversight continued; agents surfaced blockers; engineering checks provided real, though narrower, feedback. |
| Rival explanations | Identity and service configuration, migration priority, incomplete acceptance coverage, prototype debt or insufficient resources. |
| Experimental implication | Hidden retention tests and CR1 supply behavioural feedback to the evaluator. The experiment does not test live deployment recovery, product selection or a human-use checkpoint cadence. |

## A.4 Measures and lexicon

**Log entries.** The knowledge-base log (`docs/reference/log.md`) was split into entries at each second-level heading that begins with a date. An entry belongs to the date in its heading. Word counts use whitespace tokenisation of the entry, heading included.

**Evidence-boundary phrases.** Each entry was searched, case-insensitively and on word boundaries, for these patterns: *unmet*; *not accepted*; *not inferred*; *is inferred*; *remain(s) open*; *remain(s) unproved*; *remain(s) unverified*; *not claimed*; *not established*; and *no* followed, within forty letters, spaces, commas, slashes or hyphens, by *is* or *was* and then *claimed* or *inferred*. Density is matches per 1,000 words.

**Reference tracing.** An entry cites the simplicity requirements if it contains an identifier from SA-001 to SA-010 or the word *simplicity*. It cites the development profile if it contains *deploymentProfile*, *development profile* or *allowPublicNetworkAccess*.

**Repository counts.** Issue and pull-request counts come from the GitHub search API on 24 October 2026 (for example, `is:pr is:merged merged:>=2026-10-15`). Document sizes are whitespace word counts of the files at commit `29c6b51`.

**Proposed measures.** *Time to first human use*: elapsed time from the start of a delivery cycle to the first time a person uses the running increment for its purpose. *Age of the last usable increment*: time since a person last verified that the deployed system could be used for its main journey. *Behaviour-changing merge ratio*: the share of merged pull requests that change what

a user can observe in the running increment. *Cost per accepted outcome*: total model, tool and cloud cost divided by the number of outcomes a human accepted in the same period. Predeclare the outcome unit, counting rule, attribution and period. With zero accepted outcomes the ratio is undefined; report the total cost and zero count.

### A.5 Consolidation step prompt

Verbatim frozen v4 prompt from execution source e8707ecc8965d3f81f5142a2603042bfca1aab15, not the amended study's approved prompt. The archived source is byte-identical, SHA-256 75f646edbe04e1e3403b092b13edd6b66615a3362e67ad49744d229cdcb94f29. Display wrapping does not alter its text. Braces mark runner parameters.

```
<!-- markdownlint-disable-file -->
# Consolidation step prompt (arm B)

Do not add or change features. Your job is to leave the code better than you found it without
    changing its behaviour.

1. Run `make check`, then read reports/quality.md (functions above complexity 10, duplicated
    blocks, lint warnings, file sizes).
2. Review the last feature's diff (`git diff feature-start feature-end`) and the hotspots
    against this checklist:
   * duplicated logic;
   * long or complex functions;
   * unclear names;
   * dead code;
   * responsibilities in the wrong place, or rules from one feature leaking into another;
   * tests that check implementation details instead of behaviour.
3. Make small, behaviour-preserving changes. Apply every change through `./tcr "<message>"`: it
    runs all tests and commits if they pass, or reverts if they fail.
4. Never modify or delete files in tests/examples/. You may add tests that pin current behaviour
     before you refactor.
5. Rewrite DESIGN.md in at most 300 words: the current structure, where each business rule lives
    , and known debts. Replace its content; do not append.
6. Stop when the quality budget is met (no function above complexity 10, duplicated lines below
    3%, zero warnings in the configured lint rules) or when your time budget ({
    consolidation_budget}) is used.
7. Finish with a summary of at most five lines: what you changed and what debt remains.
```

### A.6 Deferred intervention measures

The study excludes intervention-rate outcomes but retains help and administrative records for eligibility and cost. These definitions are proposed future measures, not collected results. HG-DAgger and ThriftyDAgger supply cross-domain measurement precedents, not coding-agent effectiveness evidence [41, 46].

For run $r$, $n_r \geq 1$ is attempted feature cycles, including assigned post-steps and failed cycles, and $h_{rj}$ qualifying corrective-help events. The full-run rates below require complete help annotation for every attempted cycle. Otherwise report annotation coverage and leave the full-run rate unknown. Observed human intervention rate would be

$$\mathrm{HIR}_r = \frac{1}{n_r} \sum_{j=1}^{n_r} \mathbf{1}\{h_{rj} > 0\}. \tag{5}$$

Event counts and active human minutes are separate; repeated help in one cycle counts once for incidence. Let $d_{rj} \in \{0, 1\}$ indicate an explicit help request or a block needing corrective human action. Then

$$\mathrm{IDR}_r = \frac{1}{n_r} \sum_{j=1}^{n_r} d_{rj}. \tag{6}$$

Requests and supplied help differ. A timeout alone is not a help request; zero attempts gives undefined rates. A future annotation protocol needs event/cycle/phase locators, coverage checks and adjudication; absent records cannot be zero. Routine launches, planned approvals and safety stops differ from corrective rescue. Low intervention must not reward silent failure or discourage necessary human judgement.

### A.7 Historical freeze provenance

The reported case-tool snapshot is HVE Core `plugins-v3.3.106`, snapshot commit 2df5013e29ebfd2ce61bcd0cf021d8c8186482eb, generated from source e9af018fe5ffbfbc72321a8bca21fb987359af08. These are reported provenance identifiers, not verification of installation date or every session's runtime.

The historical v4 execution source was e8707ecc8965d3f81f5142a2603042bfca1aab15, with manifest-only commit 03c50856838b539780d57216af2a4cf6744187d7. Its attached osf-v4-2026-10-31.tar.gz checksum was 1a42fac40a4f1aa82df59c8c3f7f8d6dce37d22502c5952c398362ac63d6bd65. The runtime image was sha256:090220b0b172ec205f1ea473d7c3a223eec5d7d619eb21acdcf15c9289287fa3. These identify historical inputs, not a freeze or approval of the five-arm amendment. The original registration at https://osf.io/ywsrv/ is owner-reported approved after submission on 28 October; the inspected v4 prospective amendment flag was false and its completion date null. New approval and actual prospective recording cannot be inferred or backdated.